# SAMS Kernel

What has run on real hardware, and how to try the rest

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For a TI-99/4A with an F18A video card and a SAMS memory card
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Everything in this project is verified in a simulator (a TMS9900, the F18A, SAMS, a keyboard, joysticks, TI's own console and disk controller ROMs, a model of the sound chip). That finds almost every programming mistake and none of the hardware's surprises. This list is the honest boundary, in the order I would walk it on a real machine or in js99er (F18A and SAMS on). Each step says what to do, what you should see, and what to tell me if you do not.

## Run on real hardware before (by Press and The Lantern Coast, the kernel's ancestors)

SAMS paging with a 1M-or-smaller card, the keyboard decode, the F18A 80-column text screen and colour roles, files through TI's disk controller, the FinalGROM "8" image format and bank switching. **Not** run on real hardware in this form: the loader as rewritten, the module system, the page pool, and everything below. js99er with a 16M SAMS has run the kernel (which found the QUIT bug in 1.3.0, fixed in 1.3.1).

## The walk

| # | Do | You should see | If not, tell me |
|---|---|---|---|
| 1 | Load `images/hello8.bin`. Type. FCTN = | a blue screen, a title bar, your typing echoed; FCTN = returns to the console's title screen with the display normal | which message (a refusal screen names its reason), a blank screen, or garbled colours |
| 2 | `images/skdiag8.bin`, CTRL T | eight rows, all **PASS**; the first shows the page count (128 for 512K, 256 for 1M, 512 / 1024 / ... / 4096 for bigger cards); a disk in drive 1 for the last row | which row failed; the step and error on the files row |
| 3 | Leave diag with FCTN = (**with a card above 1M if you have one**) | the console title screen; the card's mapping must be off | a hang or blank screen at the reset |
| 4 | In diag, press keys and chords | each shows its code and modifiers; CTRL 3 cycles the colour schemes | a chord with the wrong code (name it) |
| 5 | `images/videomodes8.bin`, keys `1-9 0 A-E`; FCTN = | fifteen screens, one for each display mode and option; each matches `docs/VIDEO-MODES.md` | which mode, and what it looked like; PICO9918 or F18A, and its firmware |
| 6 | `images/music8.bin`: `1`, SPACE, SPACE, `S`, `2`, `B`, `W` | the tune (three voices and a beat, about 16 s, looping), silent while paused and then the same note, an A (440 Hz), a beep, and the tune playing through a two-second "work" loop | the tune is too fast or slow, a wrong note, a hiss or click, or the noise voice too loud (a PAL console needs `"fps": 50`) |
| 7 | `images/joystick8.bin`, each joystick in each direction, and fire | the matching label goes bold; the fire count rises once a press; `1`, `2`, `3` make joystick 1, 2 or both type keys (the "Keys typed" count and the last code change; UP, DOWN, LEFT, RIGHT, ENTER are `C5 D8 D3 C4 0D`) | the wrong direction or joystick, a key that sticks, ghosting with a key held |
| 8 | `images/diskbrowser8.bin` with a disk in DSK1 | the catalog; ENTER on a text file shows it; scroll; FCTN 9 back | a hang, a wrong size or type, a garbled name |
| 9 | Several files: build `tests/apps/files2` (`python3 tools/skbuild.py tests/apps/files2`) and make a disk from two text files named `src` and `src2` (`python3 tools/skdisk.py d.dsk src src2`), then load `build/files2/files28.bin` with that disk in DSK1 | nothing on the screen; afterwards the disk holds `DST`, a copy of `SRC` (check it in your usual disk manager) | the disk controller or TIPI in use; whether `DST` is complete |
| 10 | `images/workbench8.bin` with `images/workbench.dsk` in DSK1: N `DSK1.HELLO/A99` ENTER, A, then R | "Assembled: 194 bytes at >C000", then the hello program; **time the assembly with a watch and tell me** (the simulator ran 1.9 million instructions); then restart (hello has no way back) and `BEEP/A99`: | the time; any error list; a hang |

| | | | |
|---|---|---|---|
| | | A, R, keys 1-8 play a scale, FCTN 9 returns to the workbench | |
| 11 | In the workbench: E on `HELLO/A99`, type a line, CTRL S, CTRL A | the edit saves and re-assembles; a mistyped mnemonic is reported with its line | an editor key that does the wrong thing |
| 12 | Cards above 1M, if you have one: step 2 again, then 6 and 10 | the page count is right and everything else as before | the first thing that differs |

## What each unverified claim rests on

| Claim | Rests on | Where it could be wrong |
|---|---|---|
| a 12-bit SAMS register is the page with its bytes swapped (even byte the low 8 bits) | js99er's `ams.js` | a real card above 1M might lay the odd byte out differently |
| joysticks are keyboard columns 6 and 7, rows fire / left / right / down / up | my knowledge of the console's scan, and the simulator's model of it | the row order, or a joystick that needs a different column select |
| three files at once through the controller's three buffers; the controller finds a file by its PAB's address | TI's own disk controller ROM, running in the simulator (it did so correctly) | TIPI, CF7+ and other devices may differ |
| the music player writes the right bytes at the right frames | tests against a model of the sound chip | how it sounds; the noise model is an approximation |
| the workbench's assembler is right | byte-identical to xas99 for every example and two probe sources | an instruction or directive the examples do not use |
| the page tests at power-up take about a second on a 16M card | 0.25 million instructions in the simulator | an estimate; I have not timed it |

## How to report

A short note per step is enough: the step number, what you saw, the machine (console, F18A or PICO9918 with its firmware, SAMS size and brand, disk controller or TIPI), and for emulators the version and settings. A photograph of a refusal screen or the self-test is the best evidence there is.